ISM_NSP4_Lecture1_25.04.2026

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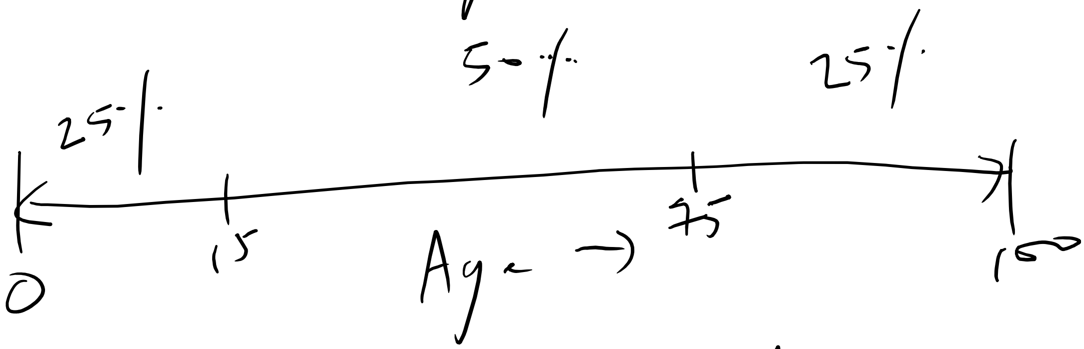
Statistics used in collecting the data?

sampling from a population so as to get a representative sample



We will draw a sample of say 1000 people → 250 children, 500 adults, 250 senior people.

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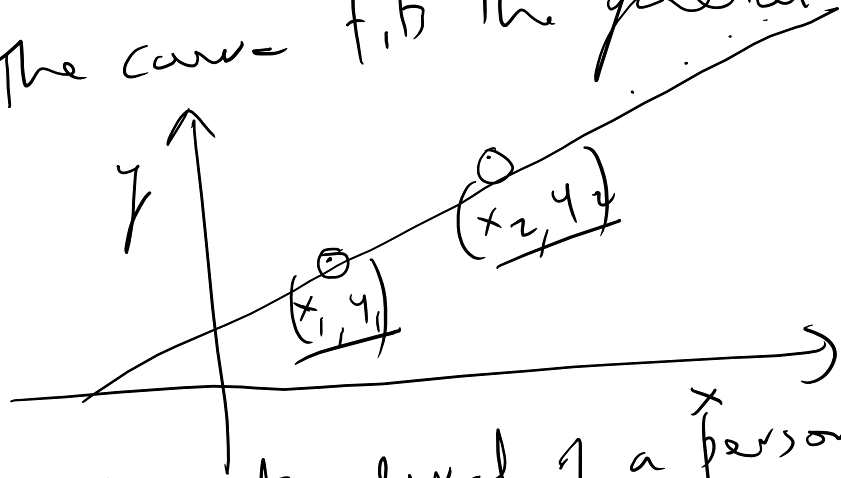
Machine learning context

(1) Collect a small sample

(2) Do curve fitting on that small sample.

(3) The curve fits the general population also.



$x_1 \rightarrow$ Education level of a person

$y_1 \rightarrow$ income of that person

What is the relationship between y & x?

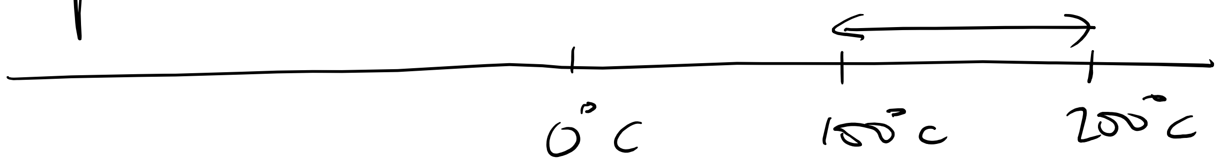
We discover a linear relationship between x and y.

As long as the <u>test data</u> is <u>drawn from the same distribution as the training data</u>, we can generalize from the training data to the <u>test data</u>

Interval scale

- difference between measurements is meaningful
- no true zero point

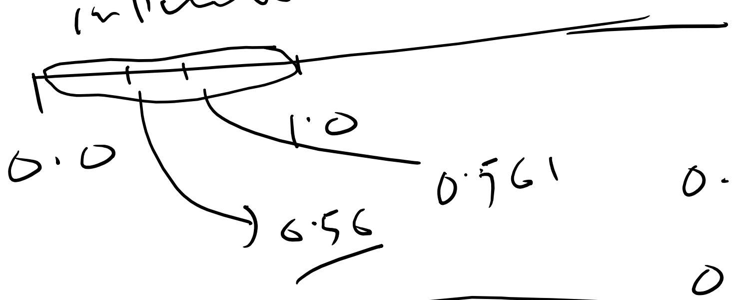
Temperature (Celsius scale)



$200°C - 100°C = 100°C$ is meaningful in the sense that 200°C is the hotter temperature

→ natural numbers
whole numbers
(countable)

infinite number of numbers here



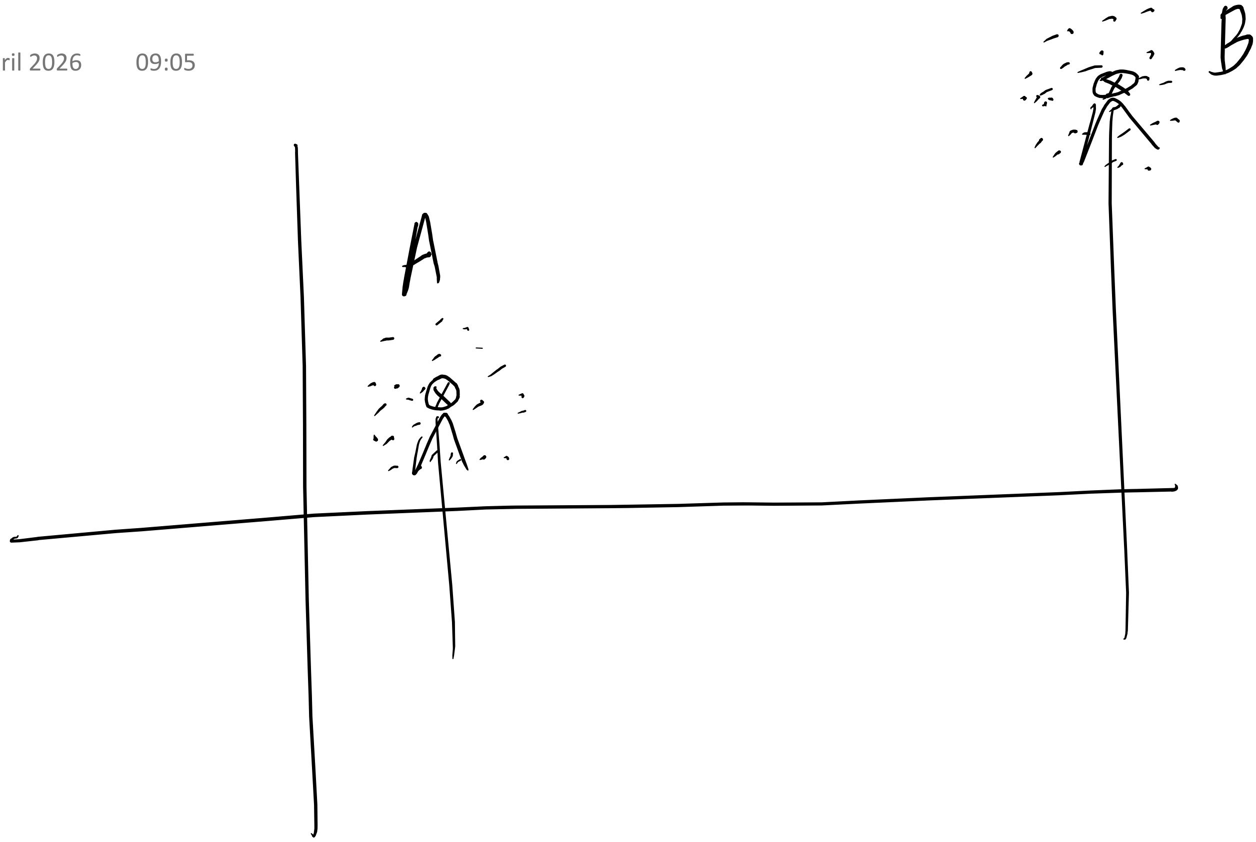
0.565821379. . .

0.5600. . . . . →1

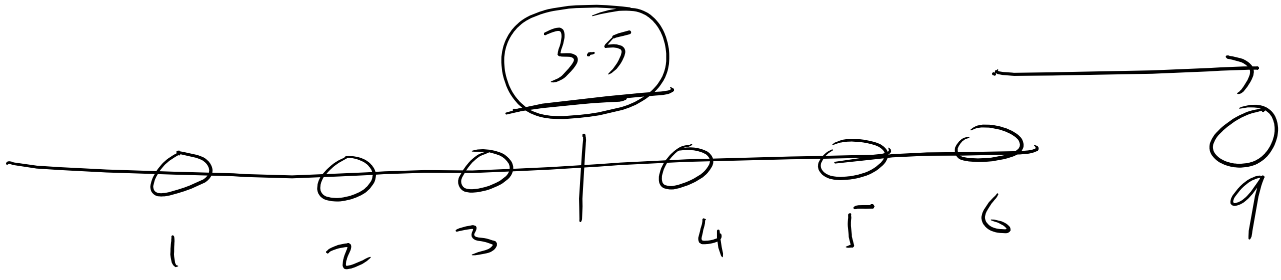
0.560. 09

0.5609999. 9

0.561


A
B

Computing mean



$$\frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

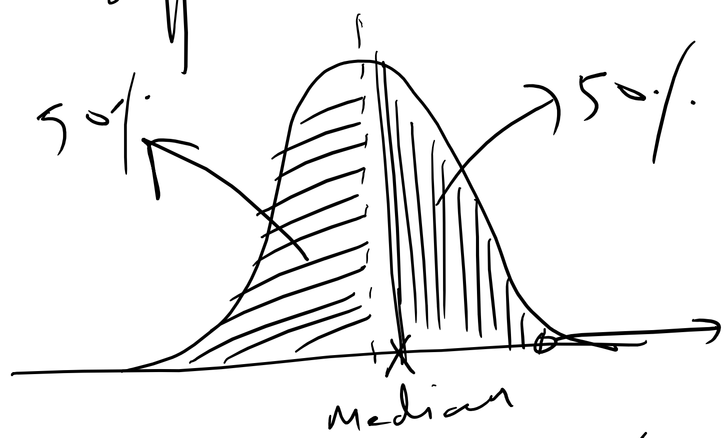
Mean is affected by outliers

$$\underset{(\text{mu})}{\mu} = \frac{x_1 + x_2 + \cdots x_N}{N}$$

all data points points

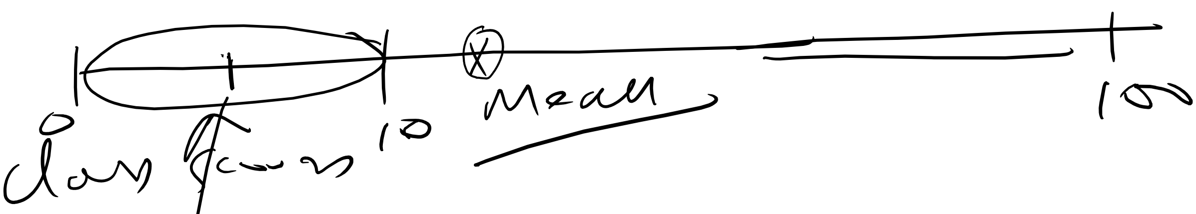
computing the mean, and therefore

affect the mean



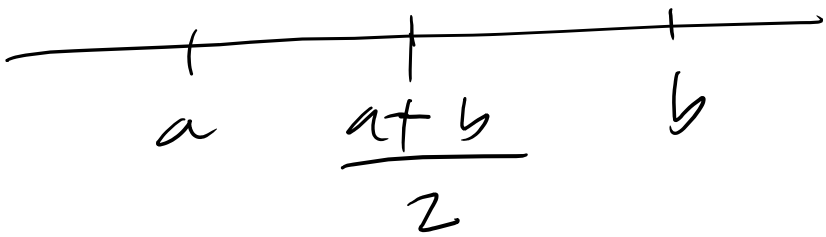
Median is less sensitive to

individual data points



Median

50 students in the class
one person gets $\frac{100}{100}$ & everyone
else gets between 0 and 10 out of
100.



$(a-5) = a'$

$(b+5) = b'$

$$\frac{a'+b'}{2} = \frac{a+b}{2}$$

But a' & b' are different data points than a & b.

$20, 20, 20, 30, 30, 30$

$$\frac{20+20+20+30+30+30}{6}$$

$$= \frac{20 \times \text{freq}_{20} + 30 \times \text{freq}_{30}}{6}$$

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$$\underbrace{\frac{x_1 + x_2}{2}}$$ vs $$\underbrace{\frac{w_1 x_1 + w_2 x_2}{w_1 + w_2}}$$

standard arithmetic mean

more general weighted mean

if $w_1 = w_2 = \frac{1}{2}$ then the weighted mean is the same as the arithmetic mean

$$\frac{\frac{1}{2}x_1 + \frac{1}{2}x_2}{\frac{1}{2} + \frac{1}{2}}$$

$$= \frac{x_1 + x_2}{2}$$

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$$\frac{X_1 + X_2 + \cdots X_N}{N} = \mu \quad - \quad (1)$$

$$\sum_{i=1}^{N} (X_i - \mu) = (X_1 - \mu) + (X_2 - \mu) + \cdots (X_N - \mu) \quad - \quad (2)$$

$X_1 + X_2 + \cdots X_N = N\mu$ from Eqn (1)

right hand side of equation (2) is

$$(X_1 + X_2 + \cdots + X_N) - N\mu$$

$$\underline{N\mu - N\mu = 0}$$

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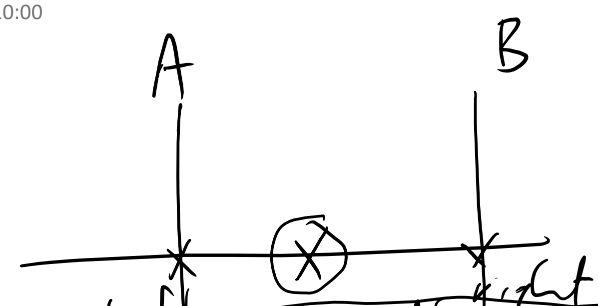

Sample mean taken from a particular sample drawn from the population

Where is the true population mean?

The population mean will lie in the interval (A, B) with high probability.

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